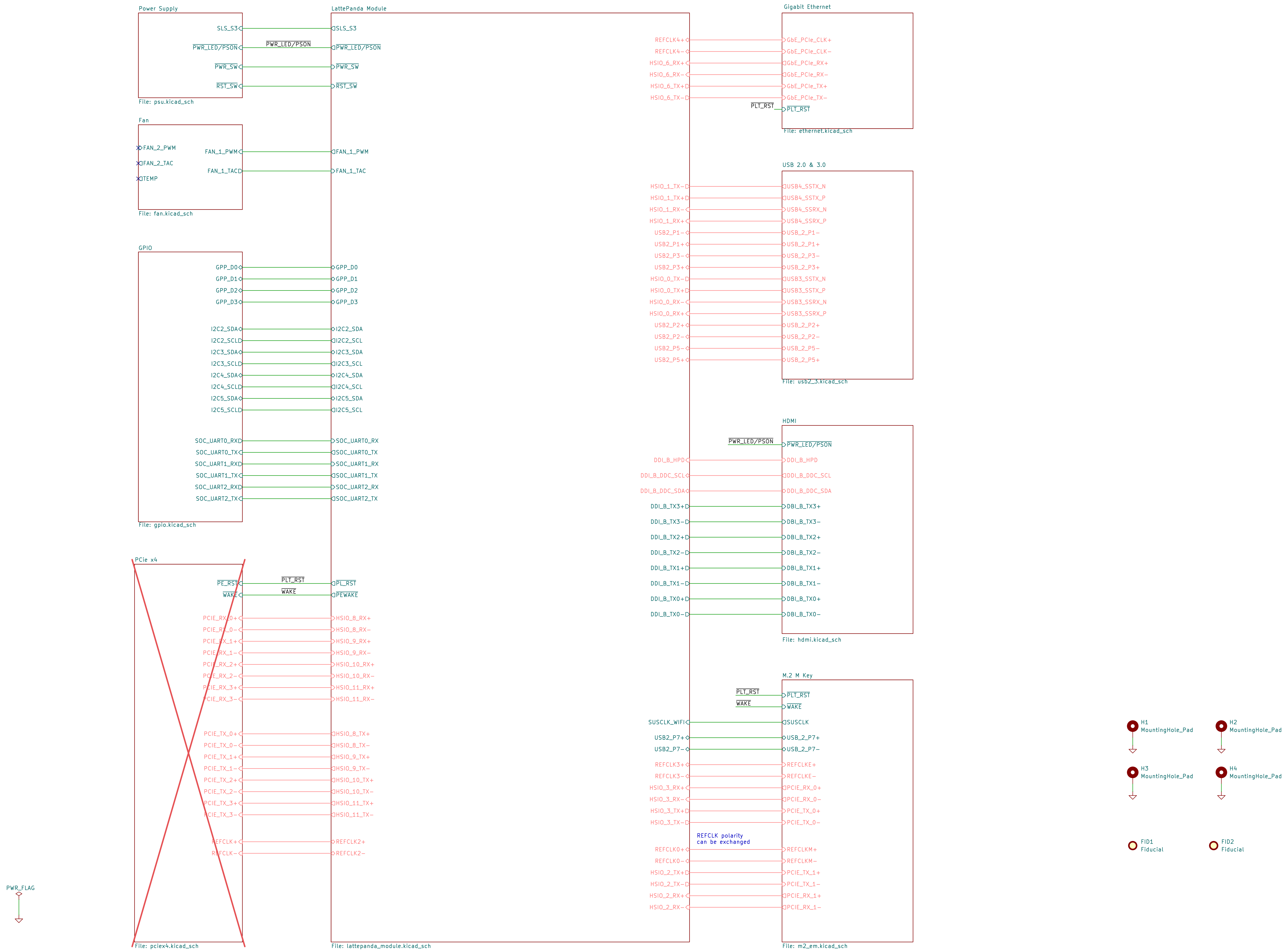


Sheet: /
File: LattePandaMU_Carrier_Custom_Board.kicad_sch

Title: github.com/tsokomalusi

Size: A2	Date: 2026-03-13	Rev: 1.0
KiCad E.D.A. 9.0.0		Id: 1/10

Sleeping State Signal

- SLS_S0:
  - S0 (Working): HIGH
  - S3 (Slpeeping): LOW
  - S4 (Hibernation): LOW
  - S5 (Soft Off): LOW
- SLS_S3:
  - S0 (Working): HIGH
  - S3 (Slpeeping): HIGH
  - S4 (Hibernation): LOW
  - S5 (Soft Off): LOW

HSIO functions are defined by the BIOS source code,
Cannot be changed in BIOS settings.

Before starting design, you should choose one of the available BIOS configurations on:
https://github.com/LattePandaTeam/LattePanda-Mu
Or contact techsupport for custom your BIOS

The default functions are defined as follows:

- **HSIO0: USB3.2 10Gbps**
- **HSIO1: USB3.2 10Gbps**
- **HSIO2: PCIe 3.0 x1**
- **HSIO3: PCIe 3.0 x1**

- **HSIO8: PCIe 3.0 x4 (Lane 0)**
- **HSIO9: PCIe 3.0 x4 (Lane 1)**
- **HSIO10: PCIe 3.0 x4 (Lane 2)**
- **HSIO11: PCIe 3.0 x4 (Lane 3)**

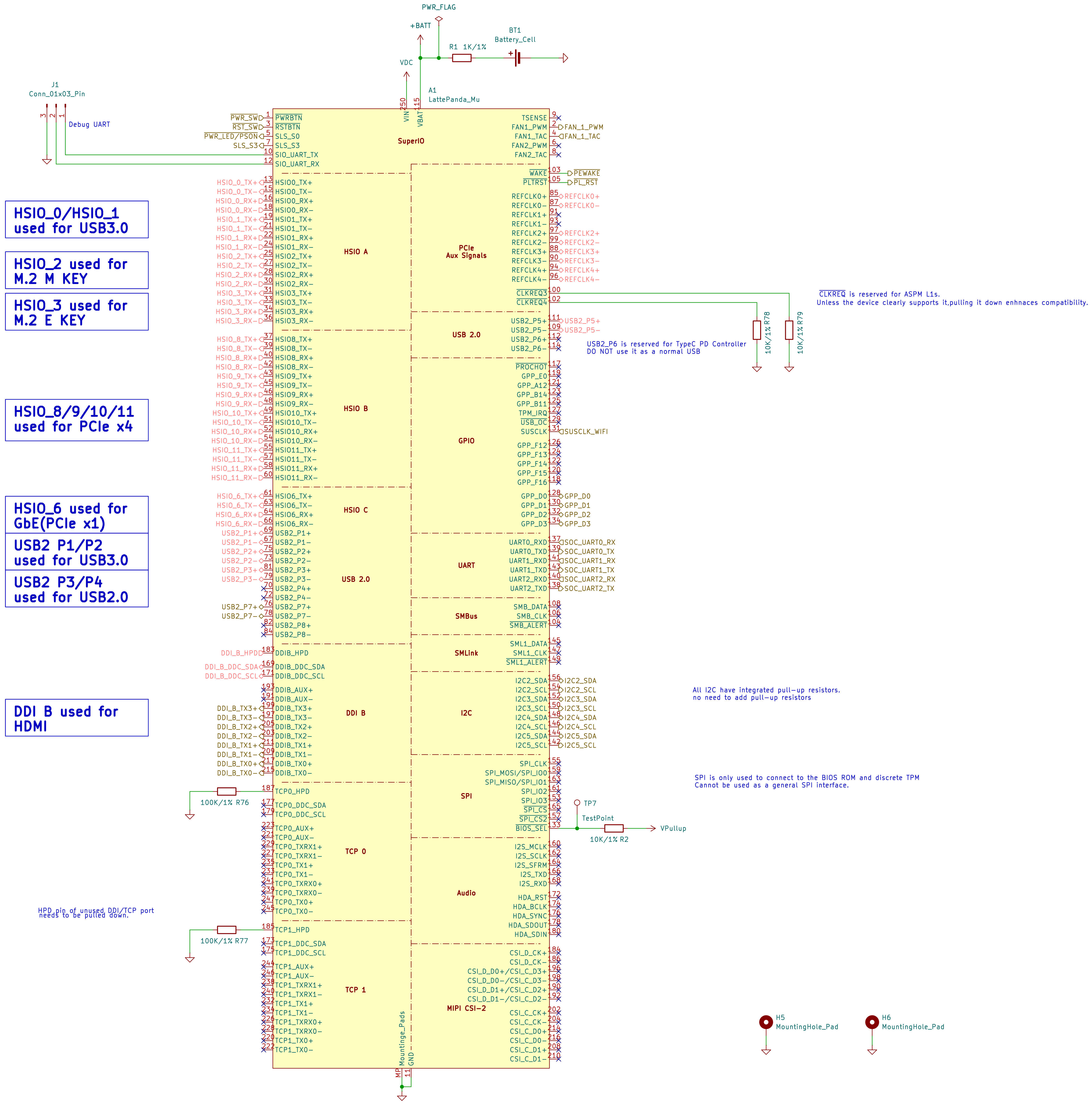
- **HSIO6: PCIe 3.0 x1**

DDI/TCP functions are defined by the BIOS source code,
Cannot be changed in BIOS settings.

Before starting design, you should choose one of the available BIOS configurations on:
https://github.com/LattePandaTeam/LattePanda-Mu

The default functions are defined as follows:

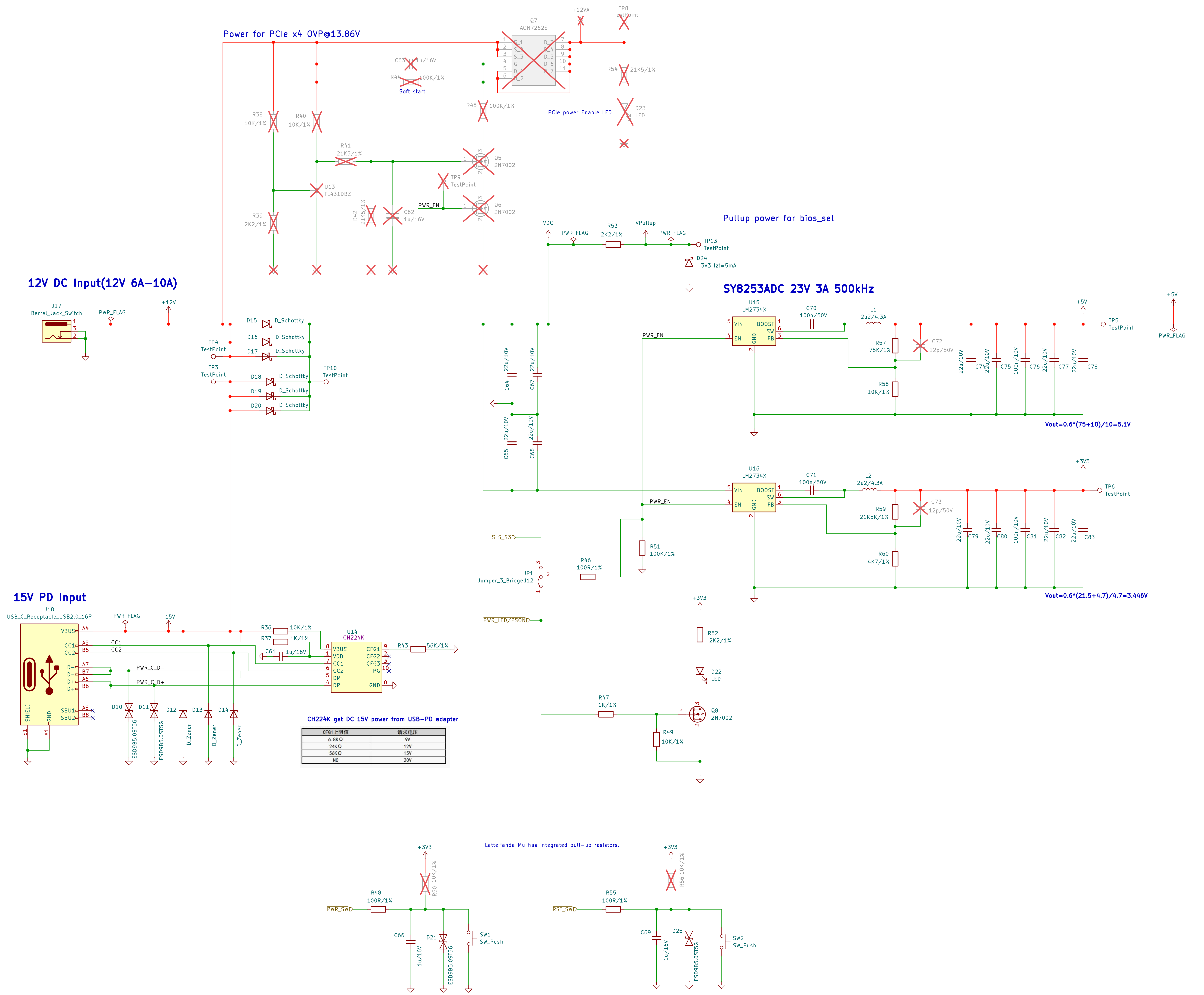
- **DDIA: eDP 1.4b (the 40Pin connector on the Mu)**
- **DDIB: HDMI 2.0**
- **TCP0: HDMI 2.0**
- **TCP1: USB TypeC (need external PD controller)**



Sheet: /LattePanda Module/
File: lattepanda_module.kicad_sch

Title:

Size: A2	Date:	Rev:
KiCad E.D.A. 9.0.0		Id: 2/10

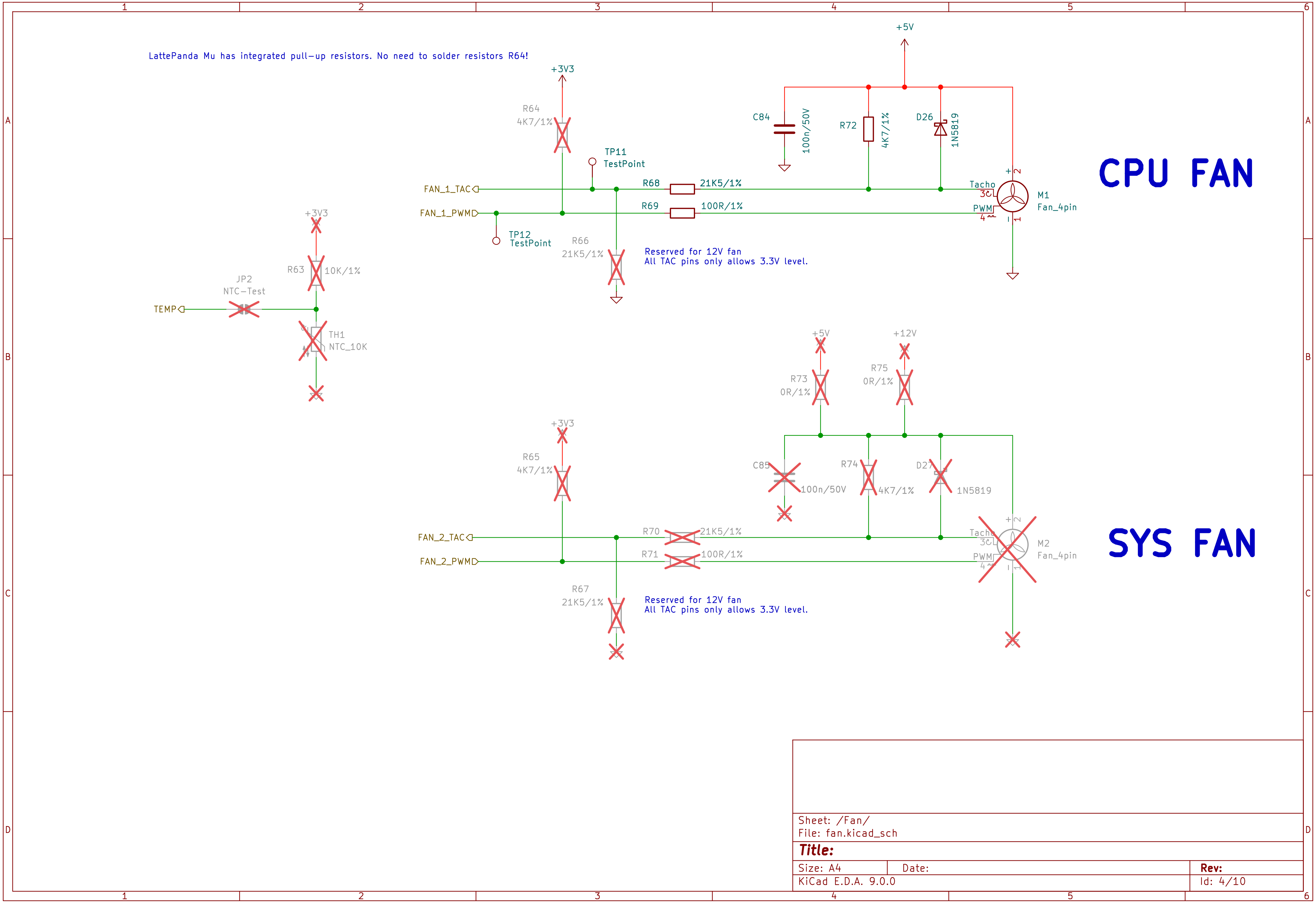


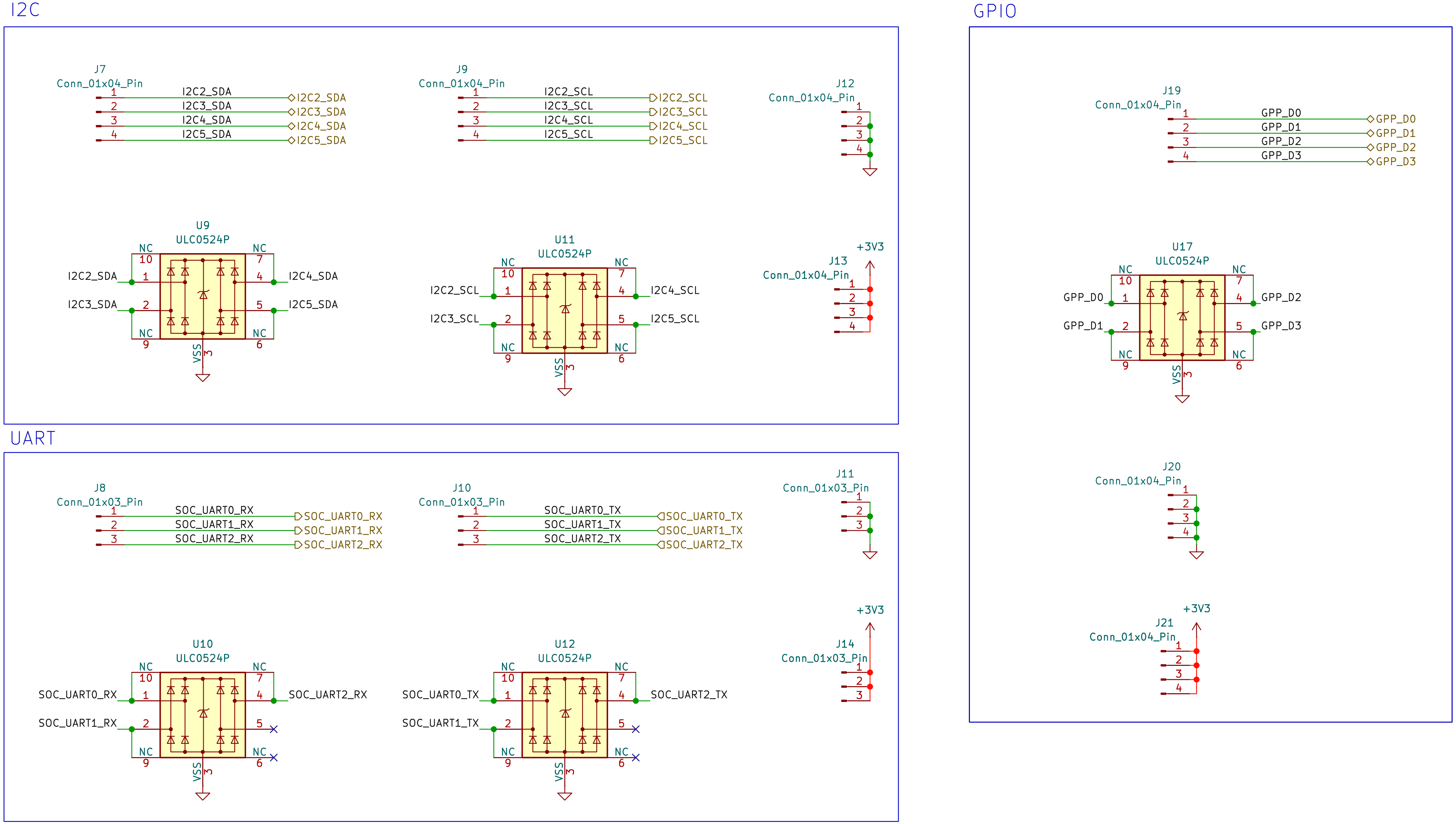
CFG1上阻值	请求电压
6.8KΩ	9V
24KΩ	12V
56KΩ	15V
NC	20V

Sheet: /Power Supply/
File: psu.kicad_sch

Title:

Size: A2 Date: Rev:
KiCad E.D.A. 9.0.0 Id: 3/10


LattePanda Mu has integrated pull-up resistors. No need to solder resistors R64!
CPU FAN
SYS FAN
+5V
+3V3
C84
100n/50V
R72
4K7/1%
D26
1N5819
R64
4K7/1%
TP11
TestPoint
FAN_1_TAC
R68
21K5/1%
Tacho
FAN_1_PWM
R69
100R/1%
PWM
M1
Fan_4pin
TP12
TestPoint
R66
21K5/1%
Reserved for 12V fan
All TAC pins only allows 3.3V level.
+3V3
R63
10K/1%
JP2
NTC-Test
TEMP
TH1
NTC_10K
+5V
+12V
R73
0R/1%
R75
0R/1%
+3V3
R65
4K7/1%
C85
100n/50V
R74
4K7/1%
D27
1N5819
FAN_2_TAC
R70
21K5/1%
Tacho
FAN_2_PWM
R71
100R/1%
PWM
M2
Fan_4pin
R67
21K5/1%
Reserved for 12V fan
All TAC pins only allows 3.3V level.
Sheet: /Fan/
File: fan.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.0
Id: 4/10

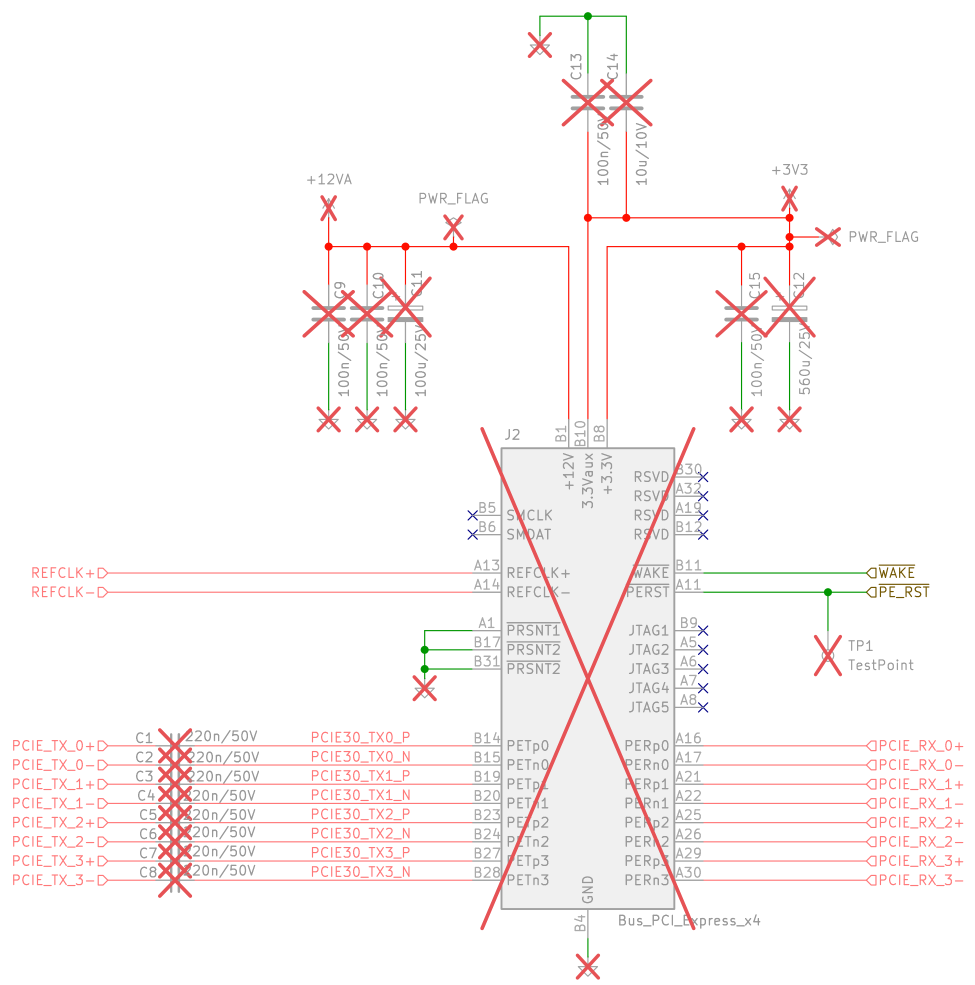


Sheet: /GPIO/
File: gpio.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 9.0.0 | Id: 5/10

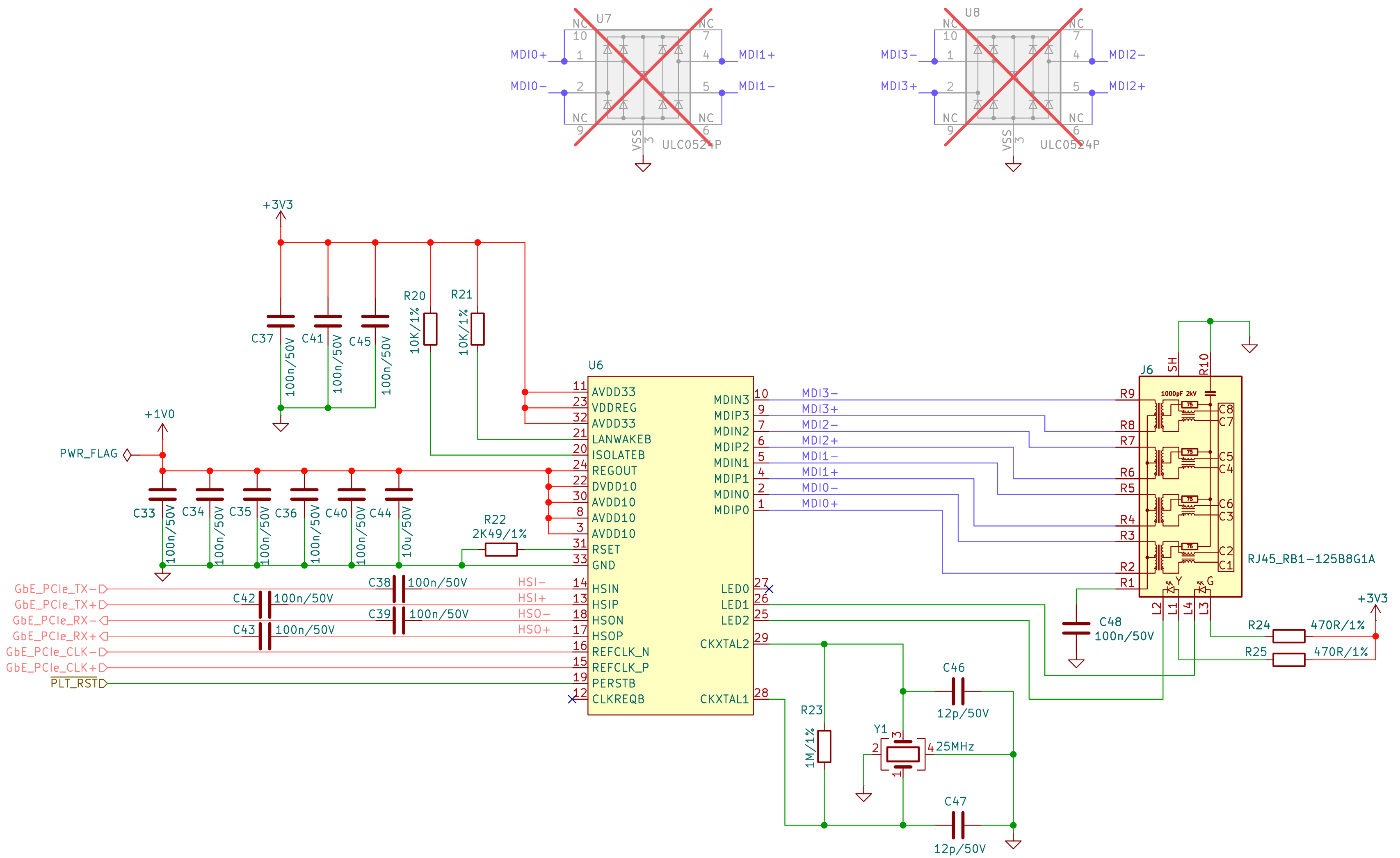


Sheet: /PCie x4/
File: pciex4.kicad_sch

Title:

Size: A4 Date: **Rev:**

KiCad E.D.A. 9.0.0 Id: 6/10



Sheet: /Gigabit Ethernet/
File: ethernet.kicad_sch

Title:

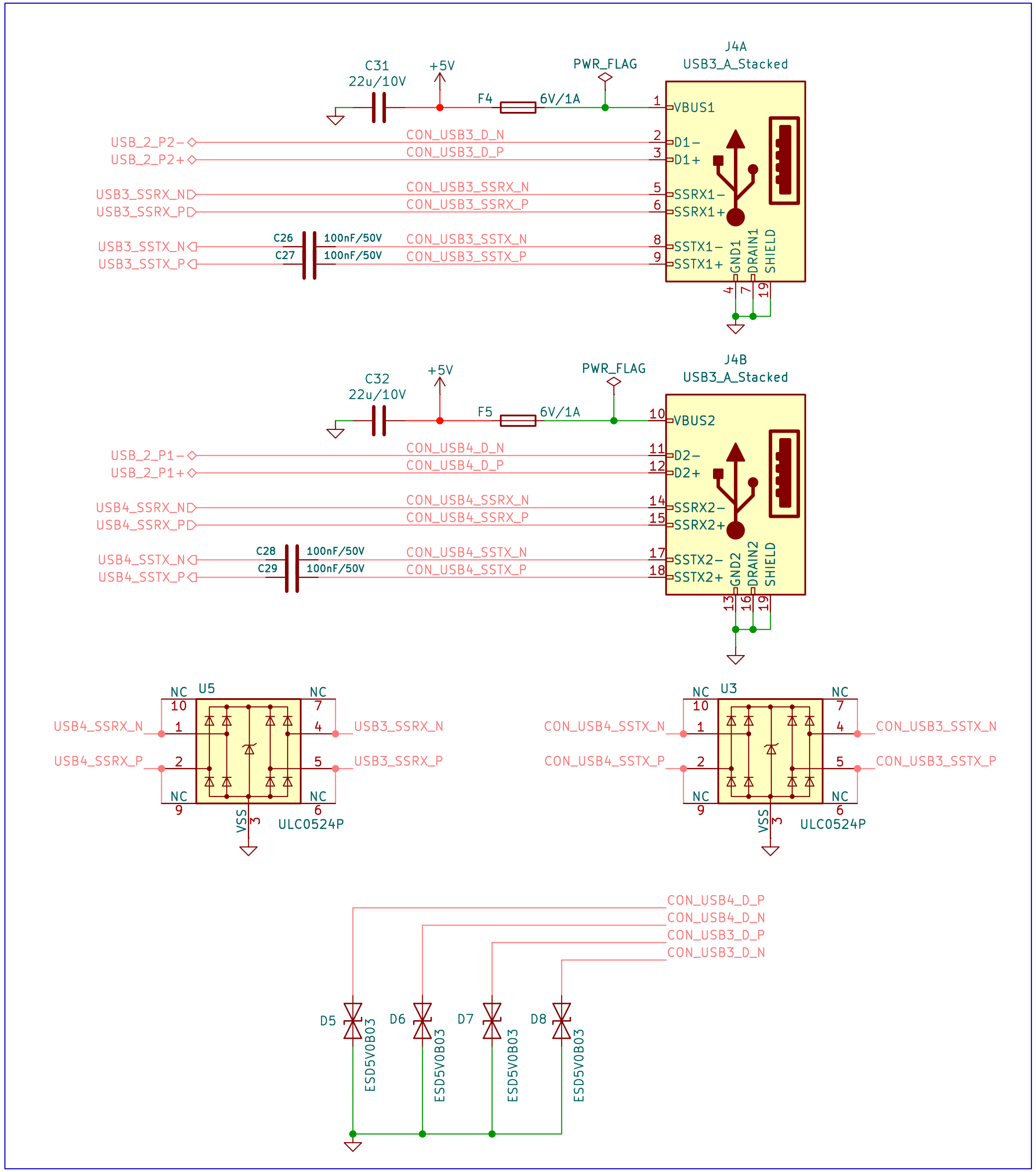
Size: A4 Date:

KiCad E.D.A. 9.0.0

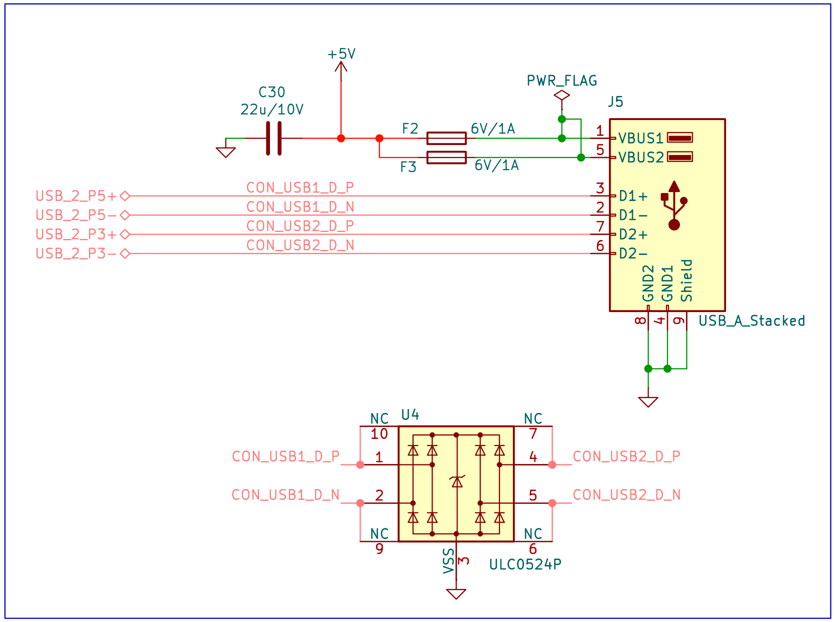
Rev:

Id: 7/10

USB3.0



USB2.0

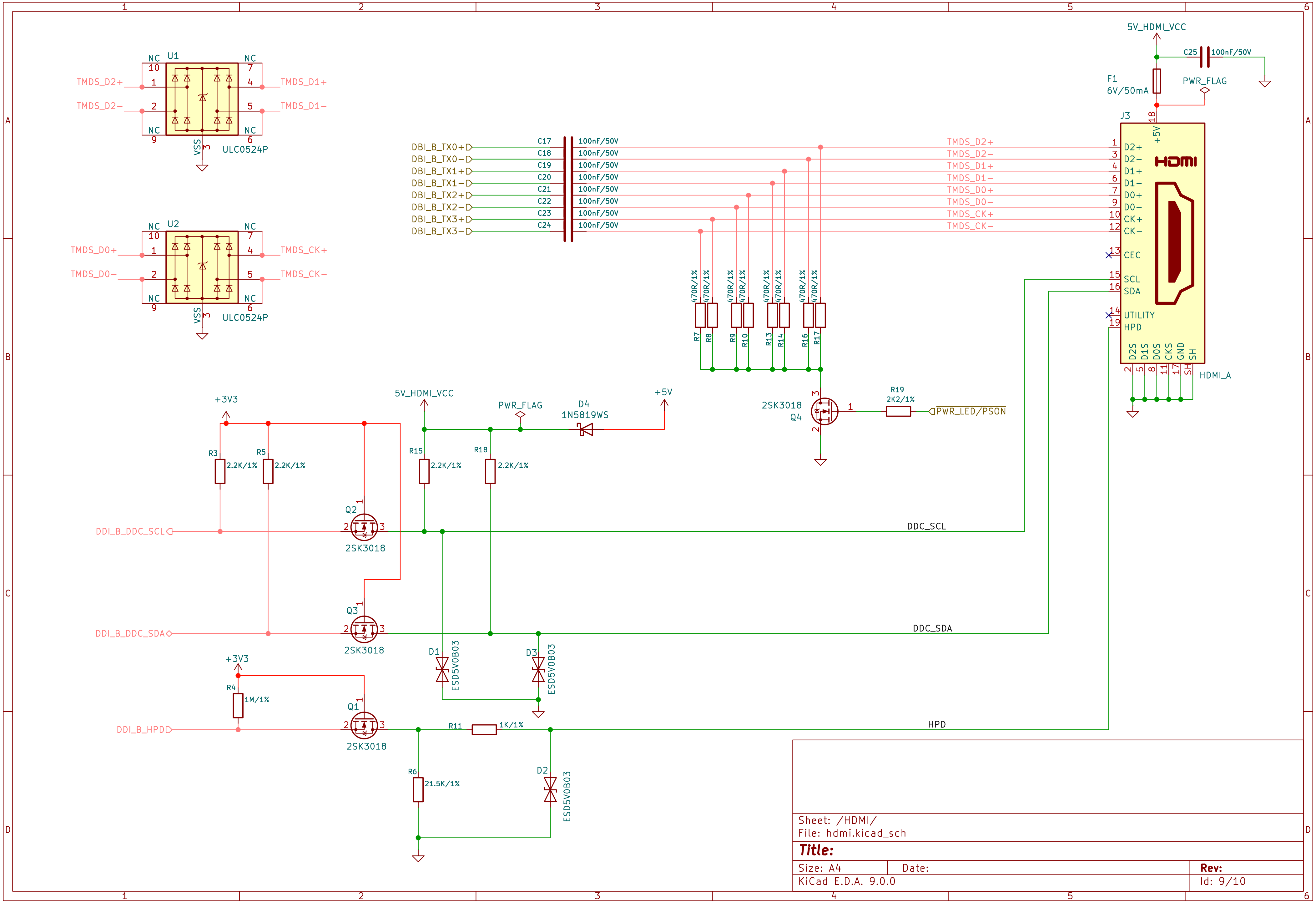


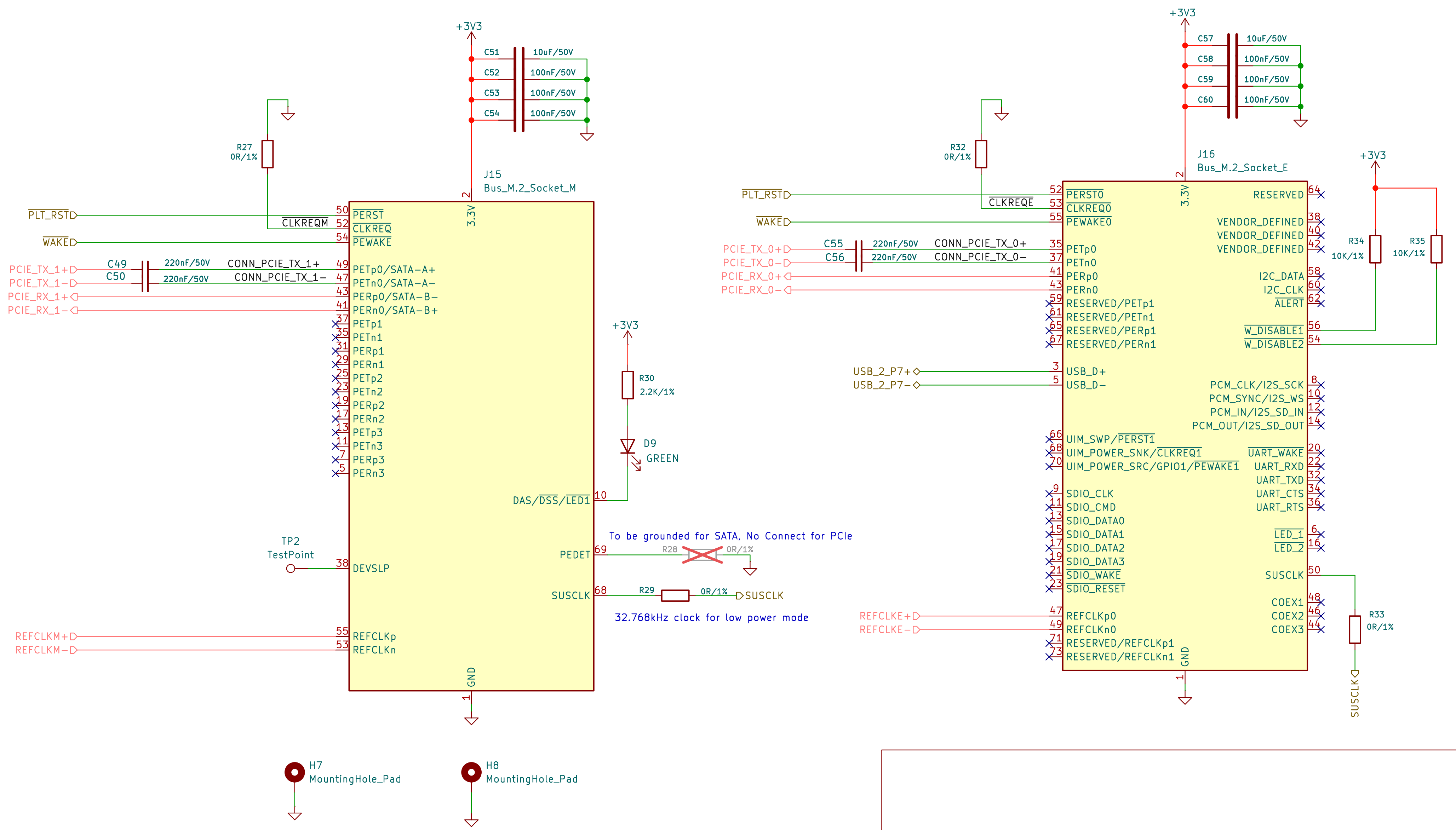
note: instructor made a mistake on D5 – D9, USB2.0 already linked to the ESD protection IC, except for USB3.x and USB4.x. Verify this later in the design.

Sheet: /USB 2.0 & 3.0/
File: usb2_3.kicad_sch

Title:

Size: A4 Date: **Rev:**
KiCad E.D.A. 9.0.0 Id: 8/10


U1
ULC0524P
U2
ULC0524P
TMDS_D2+
TMDS_D2−
TMDS_D1+
TMDS_D1−
TMDS_D0+
TMDS_D0−
TMDS_CK+
TMDS_CK−
NC
VSS
DBI_B_TX0+
DBI_B_TX0−
DBI_B_TX1+
DBI_B_TX1−
DBI_B_TX2+
DBI_B_TX2−
DBI_B_TX3+
DBI_B_TX3−
C17 C18 C19 C20 C21 C22 C23 C24
100nF/50V
R7 R8 R9 R10 R13 R14 R16 R17
470R/1%
5V_HDMI_VCC
C25
100nF/50V
F1
6V/50mA
PWR_FLAG
J3
HDMI_A
D2+ D2− D1+ D1− D0+ D0− CK+ CK− CEC SCL SDA UTILITY HPD
+5V
D2S D1S D0S CKS GND SH
2SK3018
Q4
R19
2K2/1%
PWR_LED/PSON
+3V3
R3
R5
2.2K/1%
R15
R18
D4
1N5819WS
+5V
Q2
Q3
Q1
DDI_B_DDC_SCL
DDI_B_DDC_SDA
DDI_B_HPD
DDC_SCL
DDC_SDA
HPD
D1
D2
D3
ESD5V0B03
R4
1M/1%
R11
1K/1%
R6
21.5K/1%
Sheet: /HDMI/
File: hdmi.kicad_sch
Title:
Size: A4
Date:
KiCad E.D.A. 9.0.0
Rev:
Id: 9/10



Sheet: /M.2 M Key/
File: m2_em.kicad_sch

Title:

Size: A4 Date: Rev:

KiCad E.D.A. 9.0.0 Id: 10/10